INCH-POUND
MIL-I-24783(SH)
15 April 1993

MILITARY SPECIFICATION

ISOLATOR, RING TYPE, PRESSURE, FOR SHIPBOARD SEWAGE SERVICE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for ring and flanged spool type isolators, installed between pipe flanges, to isolate pressure and vacuum gauges from clogging by sediment and solids in shipboard sewage systems.

1.2 Classification. Isolators are of the following types and sizes as specified (see 6.2).

Type A - Ring, in sizes 1-inch nominal pipe size (nps), and larger.
Type B - Flanged spool, in sizes 1/2-inch to 4-inch nps.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 05Q42, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-C-390 - Copper Alloy Castings (Including Cast Bar).
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-G-1149 - Gasket Materials, Synthetic Rubber, 50 and 60 Durometer Hardness.
MIL-S-1222 - Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-C-15726 - Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire.
MS16142 - Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.
MIL-P-16789 - Pumps (Including Prime Movers and Support Items); Packaging of.
MS18229 - Plug for "O" Ring Gasket.
MIL-G-18997 - Gauge, Pressure, Dial Indicating.
MIL-F-20042 - Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-R-83248 - Rubber Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-45662 - Calibration Systems Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE INC. (ANSI)
B40.1 - Gauges - Pressure Indicating Dial Type - Elastic Element. (DoD adopted)
B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay).

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 61 - Standard Specification for Steam or Valve Bronze Castings. (DoD adopted).
B 584 - Standard Specification for Copper Alloy Sand Castings for General Applications.
F 467 - Standard Specification for Nonferrous Nuts for General Use. (DoD adopted)
F 468 - Standard Specification for Nonferrous Bolts, Hex Cap Screws, and Studs for General Use. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.5.

3.2 Material. The material for the pressure gauge isolator shall be as specified herein. When a definite material is not specified, the material selected shall be the best commercial marine grade for the purpose intended. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Welding and brazing. Welding shall be in accordance with MIL-STD-278. Silver-brazing shall be in accordance with NAVSEA 0900-LP-001-7000.

3.2.2 Castings. Castings shall not be plugged, brazed, or impregnated or otherwise repaired (see 6.6).

3.2.3 Mercury prohibition. Components and parts furnished by the contractor shall be free of mercury contamination and free of functional mercury. During the manufacturing processes, checks, examinations, and tests, the product being offered for acceptance shall not come in direct contact with mercury, any of its compounds, or with any mercury containing device, such as gauges and thermometers. The product, when shipped, shall be free from mercury contamination (see 6.3).

3.2.4 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Construction. The ring type isolator (type A) and flanged spool isolator (type B) shall be of the design and construction as specified in 3.3.1 through 3.3.12 (see 6.3).

3.3.1 General requirements. Isolators with attached gauges will be installed between pipe flanges to monitor the pressure or vacuum in the pipe. The isolators shall function in vertical or horizontal pipe runs. The isolator shall be provided with an internal cylindrical flexible elastomeric element isolating the fluid in the pipe from the sensing fluid to the gauge. The isolator shall not cause any obstruction to fluid internal passage through the pipe. Means shall be provided to isolate the gauge or remove the gauge for repair without interrupting the flow through the pipe and without loss of the sensing fluid from the isolator assembly.

3.3.2 Isolator and gauge set. Where an isolator and gauge set is specified, isolators shall be furnished complete with mounted gauges, charged with sensing fluid, and calibrated (see 6.2).

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3.3.3 Isolator. The isolator assembly shall include a cylindrical flexible elastomeric element internally mounted in the isolator body, a gauge cutout valve, extensions, and adapters for gauge mounting, and other parts, as required by the design. The isolator assembly shall be suitable for mounting between bronze flanges in accordance with MIL-F-20042, of the size specified (see 1.2). System fluid flow passage through the isolator shall equal the nominal size of the isolator with a tolerance of plus 0.062 inches minus zero inches. There shall be no dead ends or crevices in the isolator flow passage to trap and cause sediment buildup. The inlet and outlet ways shall be chamfered to provide a smooth transition from the isolator bore to the piping system mounting flange. Isolator body material shall be tin bronze alloys C90300 or C92200 in accordance with ASTM B 61 or ASTM B 584, or QQ-C-390. Plates and shapes shall be copper alloys to QQ-N-281, or MIL-C-15726, or nickel-copper-aluminum alloy to QQ-N-286. Side faces shall be flat and machine finished as specified below (see 3.3.3.1). The isolator shall be constructed so that the elastomeric element is not compressed during installation in the piping system when pipe flanges are tightened about the isolator.

3.3.3.1 Ring isolator (type A). The isolator overall thickness shall be 2.000 ± 0.015 inches and the nominal outside diameter shall be equal to the MIL-F-20042 flange diameter. Where required by the contractor's standard construction, lugs or bosses may project outside the nominal diameter. The isolator body shall be drilled for through bolting in accordance with MIL-F-20042, class 150.

3.3.3.2 Flanged spool isolator (type B). Flange thickness, outside diameter and drilling for bolting shall be in accordance with MIL-F-20042, class 150. Isolator flange face to face dimension shall be 5.375 ± 0.062 inches.

3.3.3.3 Machine finish. Isolator faces in contact with the piping system mounting flange gasket shall have a machine surface finish in accordance with ANSI B46.1 with a circular lay of 500 to 1000 roughness height rating (RHR) or a spiral of 125 to 500 RHR, by machining 30 to 80 serrations of uniform depth per inch of face width. Where the face cannot be turned and tool marks run across the face, the surface finish shall be a maximum of 500 RHR.

3.3.4 Isolator elastomeric element. The isolator elastomeric element shall be constructed of nitrile rubber (Buna-N) and provide a smooth unobstructed flow passage through the isolator. The element shall not protrude into the flow passage more than 0.062 inches. The elastomeric element shall be contained by the isolator body, and shall be retained by mechanical means with sufficient force to make the elastomeric element joint leakproof without the use of adhesives or sealants. The elastomeric element shall not contact the piping system mounting flange or the full face mounting flange gasket.

3.3.5 Threaded fasteners. Fastener material shall be nickel-copper, grades 400 or 405 in accordance with MIL-S-1222, ASTM F 467 or ASTM F 468 or nickel-copper-aluminum grade 500 in accordance with MIL-S-1222.

3.3.6 Pressure sensing fluid. The sensing fluid contained by the isolator shall be a 50 percent ethylene glycol and water mixture, in accordance with MIL-A-46153.

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3.3.7 Fill connection. A fill connection and plug in accordance with MS16142 and MS18229 shall be provided opposite the gauge connection for filling the isolator and gauge set with pressure sensing fluid.

3.3.8 Gauge isolation valve. A cutout valve shall be provided internal to the isolator body or in the extension connections to the gauge to permit gauge removal without loss of pressure sensing fluid from isolator body cavity and without entrance of air into the body cavity.

3.3.9 Gauge connection. Connectors and adapters as required by the isolator and gauge assembly shall not extend the gauge a distance greater than 3 inches from the bottom or back of the gauge case to the top of the isolator body. Piping threaded connections shall be straight thread 9/16-18UNF-3A. Adapter fittings, O-ring unions, connectors, and special fittings shall be 90-10 copper nickel alloy in accordance with MIL-C-15726. Plugs and bosses shall be in accordance with MS16142 and MS18229. Bite-type fittings shall not be used.

3.3.10 Gauge. Gauges shall be in accordance with MIL-G-18997 as follows:

(a) Design - S.
(b) Type of pressure - As specified (see 6.2).
(c) Dial size and case design - 2.
(d) Elastic element - N.
(e) Elastic element style - N.
(f) Dial range - As specified (see 6.2).
(g) Dial color - W.
(h) Pressure connection - R.
(i) Connection location - As specified, A or O (see 3.3.10.1 and 6.2).
(j) Liquid fill - S.
(k) Vibration category - C.
(l) Cleanliness - G.

3.3.10.1 Gauge connection location. For ease of observation from system operating area, back connected, connection location O, gauges (classification S____2NN___WROSCG) shall be furnished for isolator and gauge sets installed in vertical pipe runs. Bottom connected gauges, connection location A, (classification S____2NN___WRASCG) shall be furnished for isolator and gauge sets installed in horizontal pipe.

3.3.11 Gasket and seal material. Gasket and seal material shall be compatible with and unaffected by system fluid (see 3.4.1). O-ring seals shall be fluorocarbon elastomer in accordance with MIL-R-83248. Sheet gaskets shall be synthetic rubber in accordance with MIL-G-1149.

3.4 Performance. Performance shall be as specified in 3.4.1 through 3.4.12.

3.4.1 System fluid. Isolators shall be suitable for shipboard sewage service. Shipboard sewage contains seawater with various amounts of human wastes, detergents, caustics, acids, oils, solvents, and other chemical agents used on board ship.

3.4.2 Pressure rating. Isolator maximum working pressure shall be 150 pounds per square inch (lb/in^2).

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3.4.3 Vacuum rating. Isolators 1/2-inch through 3-inch nominal size, in addition to operating under pressure (see 3.4.2) shall operate at a maximum vacuum of 20 inches mercury (hg).

3.4.4 Temperature. The isolator assembly, not including the pressure gauge, maximum operating temperature shall be 150 degrees Fahrenheit (°F).

3.4.5 Structural soundness. The isolator body assembly bolted between two flanges shall withstand an internal hydrostatic test pressure of one and one-half times the maximum pressure rating (see 3.4.2) with no leakage, weeping, or sweating through boundary material or joints, and with no structural distortion (see 4.7.2).

3.4.6 Accuracy. The accuracy of the isolator and gauge set shall be within plus or minus 5 percent of span of gauge (see 4.7.3).

3.4.7 Repeatability. The repeatability of the isolator and gauge set shall be within plus or minus 1-1/2 percent of span (see 4.7.4).

3.4.8 Extreme temperature operation. The accuracy of the isolator and gauge set operating at 145°F ambient temperature and operating at to 40°F ambient temperature shall be within plus or minus double the permitted deviation specified in 3.4.6 (see 4.7.5).

3.4.9 Installation. Compressing the isolator and gauge set between pipe flanges using normal installation practice with threaded fasteners torqued to make a leakproof joint shall not affect the gauge reading or cause a deviation from zero reading of the unpressurized gauge (see 4.7.6).

3.4.10 Vibration. The isolator and gauge set shall withstand the type I vibration requirements of MIL-STD-167-1 (see 4.7.7).

3.4.11 Shock. The isolator and gauge set shall withstand the shock requirements of grade A, hull mounted, class I, type A of MIL-S-901 (see 4.7.8).

3.4.12 Pressure cycling. The isolator shall withstand the pressure cycling requirements specified in 4.7.9 and shall show no evidence of improper operation, failure, damage, leakage of pressure sensing fluid, or leakage of system fluid. The accuracy at completion of the cycling test of the isolator and gauge set shall be within plus or minus 6 percent of span.

3.5 Identification and marking. Identification and marking of isolator and gauge assembly parts shall be as specified in 3.5.1 through 3.5.3.

3.5.1 Isolators - Each isolator shall be marked by electro-chemical etch, vibro-tool etch, or stamping with the following:

(a) Manufacturer's name or trademark.
(b) Isolator nps pipe size.
(c) Isolator pressure rating in lb/in^2.
(d) Isolator body material.
(e) Allowance parts list (APL) or National stock number (NSN) (if known).

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3.5.2 Adapters and fittings. Each adapter or special type fitting shall be marked by electro-chemical etch, vibro-tool etch, or stamping with the following:

(a) Manufacturer's name or trademark.
(b) NPS pipe size.
(c) Pressure rating.
(d) Material.
(e) Allowance parts list (APL) or National stock number (NSN) (if known).

3.6 Information plates.

3.6.1 Caution. Isolator and gauge sets (see 3.3.2) shall be provided with information plates in accordance with types A, B, C, E, F, or H of MIL-P-15024 and MIL-P-15024/5. The plate shall display the following caution:

REMOVING OR LOOSENING GAUGE WILL CAUSE LOSS OF SENSING FLUID AND GAUGE READING ERROR.

3.6.2 Maintenance. Maintenance procedures shall be provided on an 8-1/2 x 11-inch paper sheet or sheets for elastomeric element and seal replacement, gauge removal and replacement and pressure sensing fluid filling.

3.7 Workmanship. Isolators shall be in accordance with the specifications herein, and shall be without deformation or malfunction, and shall be clean and free of cracks and burrs. Surfaces of castings, forgings, molded parts, stampings, machined, and welded parts shall be clean and free from sand, dirt, sharp edges, scales, flux, and other harmful or extraneous materials. These surfaces shall also be free of defects such as cracks, porosity, under cuts, voids, and gaps. External surfaces shall be smooth and edges shall be either beveled or rounded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to specified requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection requirements set forth in this specification shall become part of the contractors overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the Contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.5).
(b) Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, inspections shall be performed in accordance with the test conditions specified herein and 4.7.

4.3.1 Environmental conditions. Unless otherwise specified herein, inspections and tests required by this specification shall be conducted at prevailing atmospheric pressure, ambient temperature of 65 to 85°F, and relative humidity of 30 to 80 percent.

4.3.2 Test fluid. Unless otherwise specified herein, all tests shall be conducted using clean fresh water.

4.3.3 Test standards. Test standards shall be as specified in 4.3.3.1 and 4.3.3.2.

4.3.3.1 Test gauge. A test gauge, in accordance with ANSI B40.1, shall be used to check the accuracy of the isolator and gauge assembly. The accuracy of the test gauge shall be within plus or minus 1/4 percent of span. The test gauge shall be maintained in accordance with MIL-STD-45662.

4.3.3.2 Other standards. Upon prior approval, another type of pressure standard may be used (see 6.6). The approved pressure standard shall be not less than four times more accurate than the isolator and gauge assembly being tested or shall operate in the local environment within one-fourth tolerance specified for the isolator and gauge assembly being tested. The approved pressure standard shall be maintained in accordance with MIL-STD-45662.

4.3.4 Gauge for test purposes. Where a first article test sample is an isolator without gauge, first article tests shall be conducted using a 2-1/2 inch-dial size pressure gauge, dial range 0 to 200 lb/in^2, or 30 inches hg - 200 lb/in^2, as applicable. The sample isolator with gauge installed shall be considered as an isolator and gauge set in the test requirements herein.

4.4 Rejection criteria. One or more failures of any test or one or more defects found in any examination shall be cause for rejection of the entire lot represented by the sample. Rejection of a lot shall require that corrective action be implemented by the contractor.

4.5 First article inspection. When required (see 3.1), first article inspection shall consist of the examinations and tests specified in table I.

4.5.1 First article inspection samples. Unless otherwise specified (see 6.2), one sample of each type and size of isolator or one sample of each type, size, and pressure range of an isolator and gauge set shall be subjected to first article inspection specified in 4.5. The first article may be a standard product-

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TABLE I. Dimensions and tolerances for fittings, gauges, bosses

ion item from the contractor's current inventory, provided the sample meets the requirements of this specification and is representative of the design, construction, materials, and manufacturing technique applicable to the remaining isolators and isolator and gauge sets to be furnished under the contract.

4.5.2 Order of tests. Examination and tests listed in table I shall be performed in the order listed (see 6.6).

TABLE I. First article inspection.

Inspection	Requirement	Test
Visual and dimensional examination	3.3.2 to 3.3.10, 3.5, 3.5.1, 3.5.2, 3.5.3, 3.6, and 3.7	4.7.1
Structural soundness 1/	3.4.5	4.7.2
Accuracy	3.4.6	4.7.3
Repeatability	3.4.7	4.7.4
High temperature	3.4.8	4.7.5
Installation tightening	3.4.9	4.7.6
Vibration	3.4.10	4.7.7
Shock	3.4.11	4.7.8
Pressure cycling	3.4.12	4.7.9
Inspection of packaging	5	4.7.10

1/ Structural soundness test shall be performed prior to installing gauges specified (see 3.3.10).

4.6 Quality conformance inspection. Quality conformance inspection shall consist of examination and tests listed in table II (see 6.3).

TABLE II. Quality conformance inspection.

Inspection	Requirement	Test
Visual and dimensional examination	3.2, 3.3, 3.5, 3.5.1, 3.5.2, 3.5.3, 3.6, 3.7	4.7.1
Structural soundness 1/	3.4.5	4.7.2
Accuracy	3.4.6	4.7.3
Inspection of packaging	5	4.7.10

1/ Structural soundness test shall be performed prior to installing gauges specified (see 3.3.10).

4.6.1 Inspection lot. All isolators and isolator and gauge sets of the same class, nominal size, and pressure range offered for delivery at the same time and manufactured within the preceding 12 months shall be considered a lot for purposes of sampling.

4.6.2 Sampling for visual and dimensional examination. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed isolators or isolator and gauge sets in accordance with table III and subject them to the examinations listed in table II. The contractor has the option of screening 100% of the lot for the defective characteristic(s) or providing a new lot which shall be examined in accordance with the sampling plan contained herein.

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The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE III. Sampling for visual and dimensional examination. 1/ 2/ 3/

Lot size	Sample size: Major defects	Sample size: Minor defects
2 - 15	5	2
16 - 25	5	3
26 - 50	5	5
51 - 90	7	5
91 - 150	11	6
151 - 280	13	7
281 - 500	16	9
501 -1200	19	11
1200 -3200	23	13

1/ All defective items must be replaced with acceptable items prior to lot acceptance.
2/ Inspect sample size until reject criteria is reached.
3/ Rejected lots may be screened and resubmitted for inspection and retest.

4.6.3 Sampling for quality conformance tests. As a minimum, the contractor shall randomly select a sample quantity of completed isolators or isolator and gauge sets in accordance with table IV and subject them to the tests listed in table II. The contractor has the option of screening 100% of the lot for the defective characteristic(s) or providing a new lot which shall be tested in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE IV. Sampling for quality conformance tests. 1/ 2/ 3/

Lot size	Sample size
2 - 15	All
16 - 280	20
281 - 1200	47
1201 - 3200 4/	53

1/ All defective items must be replaced with acceptable items prior to lot acceptance.
2/ Inspect sample size until reject criteria is reached.
3/ Rejected lots may be screened and resubmitted for inspection and retest.

4.6.4 Order of tests. Examinations and tests specified in table II shall be performed in the order listed.

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4.6.5 Sampling for inspection of packaging. For purposes of inspection, a completed package shall be considered a unit of product for the packaging examination and a completed pack prepared for shipment shall be considered a unit of product for the packing examination.

4.7 Methods of inspection.

4.7.1 Visual and dimensional examination. Each sample item selected shall be visually and dimensionally examined to determine conformance with this specification, workmanship requirements, and applicable drawings. Major and minor defects shall be classified as shown in table V. Any sample failing to conform to the visual and dimensional examination shall be rejected.

TABLE V. Classification of defects.

Major	
101	Type and class not as specified.
102	Gauge pressure range not as specified.
103	Dimensions not as specified.
104	Material not as specified.
105	Cleanliness, dirt, chips
106	Improper assembly, parts missing
107	Isolator elastomeric element, gouged, cut, or improperly installed.
108	Sensing fluid leakage.
109	Gauge case fluid leakage.
110	Workmanship, machine finish.
Minor	
201	Identification marking missing, incorrect, or illegible.
202	Packaging nonconforming.

4.7.2 Structural soundness. The isolator body, filled with sensing fluid and fitted with a test gauge of suitable range or with gauge connection plugged, shall be hydrostatically tested to an internal pressure one and one-half times the pressure rating (see 3.4.2), to determine compliance with 3.4.5. The test pressure shall be maintained for 5 minutes.

4.7.3 Accuracy. The isolator and gauge set shall be tested for accuracy to determine compliance with 3.4.6. Accuracy shall be determined as specified in 4.7.3.1 and 4.7.3.2.

4.7.3.1 Precycling. The precycling procedure shall be performed before the start of the accuracy test to remove friction in the gauge movement. The pressure gauge shall be cycled over the entire span, but not greater than rated pressure (see 3.4.2), by slowly increasing then decreasing the applied pressure three times. The vacuum gauge shall be cycled over the entire span by slowly increasing then decreasing the applied vacuum three times.

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temp. or missing prior to plating

4.7.3.2 Accuracy procedure. The accuracy test shall consist of a set of readings at five equally spaced points over the entire span for one cycle. Where the dial span exceeds the isolator rated pressure (see 3.4.2) the test shall be conducted up to rated pressure. This cycle shall consist of reading the five equally spaced points while increasing the pressure throughout the span, then reading the same five equally spaced points in the reverse order while decreasing the pressure throughout the span. These five equally spaced points shall include the high and low limits of the span. Readings shall be taken after the pressure gauge is lightly tapped in the center of the dial. The pressure gauge indications, after tapping, for each of the five equally spaced points read while both increasing and decreasing the pressure, shall meet the accuracy requirements specified in 3.4.6.

4.7.4 Repeatability. The isolator and gauge set shall be tested for repeatability to determine compliance with 3.4.7. The accuracy test as specified in 4.7.3 shall be performed two additional times. The difference between any two readings, after tapping, at the same pressure, approached from the same direction, taken during the accuracy and repeatability tests shall be referred to as repeatability. The data taken during the accuracy repeatability test shall meet the repeatability requirements specified in 3.4.7.

4.7.5 Temperature. The isolator and gauge set shall be tested at high ambient temperatures and low ambient temperatures to determine compliance with 3.4.8

4.7.5.1 High temperature. The isolator and gauge set shall be placed in a temperature test chamber and shall be pressurized to the midspan of the gauge. The temperature in the test chamber shall be brought to 145 ± 5°F and allowed to stabilize. After the temperature in the test chamber has stabilized at 145 ± 5°F, the isolator and gauge assembly shall remain in the test chamber for at least 4 hours while pressurized at midspan. After this minimum 4-hour period, a reference measurement shall be performed while isolator and gauge assembly remains at 145 ± 5°F inside the test chamber. The isolator and gauge assembly shall then be unpressurized and removed from the test chamber, and allowed to stabilize at the ambient conditions (see 4.3) for at least 4 hours. A second reference measurement shall then be performed. The isolator and gauge assembly shall meet the requirements specified in 3.4.8.

4.7.5.2 Low temperature. The isolator and gauge set shall be subjected to the same test procedure as 4.7.5.1 except that the test chamber shall be set for a temperature of 40 ± 5°F. The isolator and gauge performance shall meet the requirements specified in 3.4.8.

4.7.6 Installation. The isolator and gauge set shall be installed between pipe flanges to determine compliance with 3.4.9. The unpressurized isolator and gauge set shall be calibrated to zero reading prior to installation, then compressed between pipe flanges as specified in 3.4.9. Deviation from zero reading of the unpressurized gauge shall be cause for rejection.

4.7.7 Vibration. The isolator and gauge set shall be vibration tested to determine compliance with 3.4.11 (see 6.3). The isolator shall be fitted with a gauge not previously subjected to any test. The isolator and gauge set shall be in operating condition filled with pressure sensing fluid and shall be pressure tested for accuracy using procedures specified in 4.7.3. The vibration test shall be performed unpressurized. After test the isolator and gauge set shall be tested

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for accuracy to determine compliance with 3.4.6. Leakage of sensing fluid or test fluid, structural damage, or permanent deformation shall also be cause for rejection.

4.7.8 Shock. The isolator and gauge set shall be shock tested to determine compliance with 3.4.10 (see 6.3). The isolator and gauge set shall be in operating condition filled with pressure sensing fluid. The shock test shall be performed unpressurized. After test the isolator and gauge set shall be pressurized 10 percent above the maximum working pressure (see 3.4.2). Leakage of sensing fluid or test fluid, structural damage, or permanent deformation shall be cause for rejection.

4.7.9 Pressure cycling. The isolator shall be pressure cycle tested to determine compliance with 3.4.12 using a gauge of suitable range. The pressure cycling test shall be performed using a pressure cycler that subjects the isolator to a smooth as practicable application and release of pressure. Hydraulic pressure cycles shall have a sinusoidal pressure output. The isolator shall be subject to the following criteria:

(a) A pressure cycle from 20 ± 4 percent to 100 percent of the pressure rating (see 3.4.2 and 3.4.3).
(b) A frequency rate of 1 ± 0.5 cycles per second (Hz) for pressure cycles only, for vacuum cycles the frequency rate of the manufacturer's test equipment shall be used.
(c) A total of 130,000 cycles each for pressure and vacuum, except where the isolator is not vacuum tested, the total pressure cycles shall be 260,000.

4.7.9.1 Acceptance procedures. The isolator shall be fitted with a gauge not previously subjected to any test, using procedures recommended by the contractor. Isolator and gauge set shall be pressure tested for accuracy up to rated pressure (see 3.4.2 and 3.4.3) using procedures specified in 4.7.3 After completion of accuracy testing the isolator body shall be disassembled and the isolator elastomeric element (see 3.3.4) shall be carefully examined for cracks, tears, cuts, or other deterioration and the condition of the elastomeric element shall be described.

4.7.10 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified (see 6.2).

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions.)

5.1 General.

5.1.1 Navy shipboard stowage fire retardant requirements.

a. Wood Unit Containers. Unless otherwise specified (see 6.2) all wood and plywood containers including laminated veneer material,

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used as unit containers shall be fire retardant treated with material conforming to MIL-L-19140 as follows:

Levels A and B	Type II - Weather resistant Category 1 - general use
Level C	Type I - Non-weather resistant Category general use

b. Fiberboard Unit Containers. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation. Isolators and isolator and gauge shall be preserved Level A or Commerical as specified (see 6.2).

5.2.1 Level A. Preservation shall be Level A in accordance with definitions listed in MIL-STD-2073-1.

5.2.2 Commercial. Commerical preservation shall be in accordance with ASTM D 3951.

5.2.3 Cushioning and wrapping materials. The use of excelsior, newspaper, shredded paper (all types), and similar hygroscopic or nonneutral materials and all types of loose fill materials for packaging applications such cushioning, fill, stuffing, and dunnage is prohibited. Cushioning or wrapping materials shall be provided to prevent item and package damage and to prevent free movement of container contents.

5.3 Packing. Packing shall be Level A, B, C, or commerical as specified (see 6.2).

5.3.1 General. Shipping containers shall contain identical quantities of identical material and shall be minimum weight and cube, similar construction, and of uniform size.

5.3.2 Level A, B, and C containers. Material preserved as specified (see 5.2) shall be packed in shipping containers, cleated plywood or nailed and locked corner boxes or covered crates, for the level packing specified (see 6.2), in accordance with MIL-STD-2073-1, appendix A, table VII. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Closure, gross weight, and waterproofing.

5.3.2.1.1 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable specification of appendix thereto except that MIL-STD-2073-1, class weather resistant fiberboard boxes shall be closed in accordance with Method V, and reinforced with nonmetallic or tape banding, and non-weather resistant fiberboard boxes shall be closed in accordance with Method I using pressure sensitive tape.

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1) Pieces 4-inch NPS and less machined directly from bars, rods or

5.3.2.1.2 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.2.1.3 Waterproofing. Unless otherwise specified (see 6.2) level A and when specified (see 6.2) level B shipping containers shall be provided with caseliners, linings, wraps, or shrouds in accordance with the waterproofing requirements of MIL-STD-1186.

5.3.3 Commerical. Material preserved as specified (see 6.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with the minimum of 3- by 4- inch nominal wood skids laid flat, or a skid- or sill- type base that will support the material and facilitate handling by mechanical handling equipment during shipment.

5.4 Marking. Marking shall be in accordance with MIL-P-16789. Packing shall also be marked as follows:

"CONTAINS FRAGILE INSTRUMENT HANDLE WITH CARE"
"SHIPPING AND STORAGE TEMPERATURE LIMITS: -10 TO 150°F".

5.5 Maintenance label plates. A copy of the maintenance label plate (see 3.6.2) shall be packed with each isolator and isolator and gauge set.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Gauge isolators covered by this specification are intended for installation in fluid systems that contain large amounts of sediment or solids. The isolators protect pressure and vacuum gauges from clogging or providing inaccurate readings. The isolator full bore design does not restrict fluid flow and the smooth interior reduces sediment buildup.

6.1.1 Gauge pressure ranges. Dial ranges of isolator and gauge sets for shipboard sewage service are generally as follows:

Pressure
0-60 lb/in^2
0-100 lb/in^2

Compound pressure
30 inches hg - 50 lb/in^2
30 inches hg - 100 lb/in^2

Vacuum pressure
30 inches hg

6.1.2 Raised face flanges. Isolators specified herein are not suitable for installation in pipe systems with raised face flanges.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of isolator required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specified issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Whether a first article inspection is required (see 3.1).
(e) Whether an isolator and gauge set is required (see 3.3.2).
(f) Type of pressure (see 3.3.10(b)).
(g) Dial range (see 3.3.10(f)).
(h) Connection location (see 3.3.10(i)).
(i) When first article inspection sampling is other than as specified (see 4.5.1).
(j) Inspection of commercial packaging (see 4.7.10).
(k) Preservation and packing levels (see 5.2).
(l) When drawings are required (see 6.3)

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.3	DI-MISC-80678	Certification/data report	10.3.1 does not apply
3.3, 3.6	DI-DRPR-81000	Product drawings and associated lists	---
4.1.1	DI-QCIC-81110	Inspection and test plan	---
4.6	DI-NDTI-80809	Test/inspection reports	---
4.7.7	DI-ENVR-80708	Shock test report	---
4.7.8	UDI-T-23762	Report, vibration testing	---
5.1	DI-PAK-80120	Preservation and packing	---

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required (see 6.2), the contracting officer should provide specific guidance to offerors whether the item should be a first article sample (see 3.1), and the number of items to be tested as specified in 4.5.1. The contracting officer should also include specific

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instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 NAVSEA approval and direction. Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures require NAVSEA approval or direction. Requests should include supporting documentation.

6.7 Subject term (key word) listing.

Chemical isolator
Diaphragm seal type isolator
Gauge isolator
Instrument isolator

Preparing activity:
Navy - SH
(Project 4730-N092)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-24783(SH)	930415

3. DOCUMENT TITLE

ISOLATOR, RING TYPE, PRESSURE, FOR SHIPBOARD SEWAGE SERVICE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
TECHNICAL POINT OF CONTACT: Mr. David Kopack, SEA 05V12	703-602-0351	332-0351
c. ADDRESS (Include Zip Code) COMMANDER, SEA 05Q42 NAVAL SEA SYSTEMS COMMAND 2531 JEFFERSON DAVIS HWY ARLINGTON, VA 05V12	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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